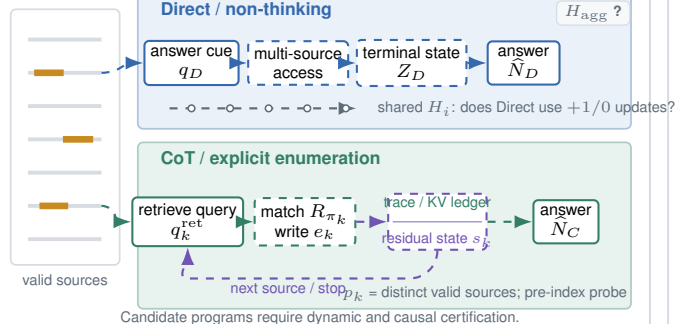


A Competing counting programs

Same sparse context; solid boxes are registered variables, dashed paths are hypotheses.

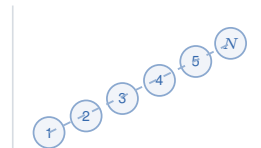
shared context



B Registered hidden states

Geometry localizes candidate count/progress states; it does (SCHEMATIC — NOT DATA)

Direct: Z_D and $H_i = h(a_i)$



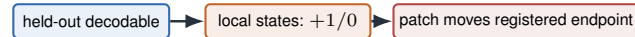
PCA(Z_D | N); H_i only predicate-before

CoT: $P_k = h(q_k^{end})$



PCA(P_k | p_k); neutral pre-index landmarks

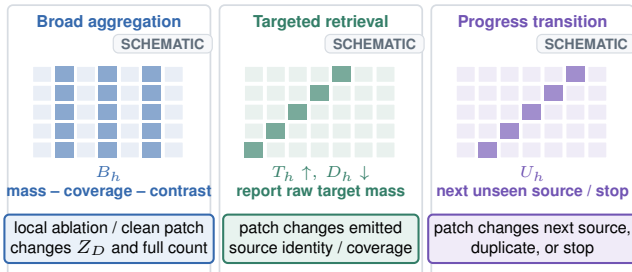
State certification



H_i, P_k : all three; terminal Z : total decoding + causal transport; geometry is descriptive.

C Operational attention roles

Attention over record spans nominates candidates; held-out output interventions certify roles.

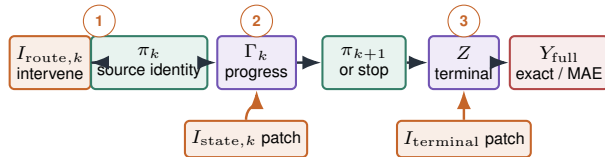


Certification: discover → freeze → held-out → local ablation → clean rescue.

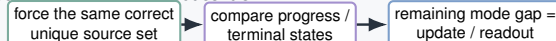
Matched controls required; routing score $\neq$ component, and update may reside in MLP/residual.

D Time-indexed causal bridge

Intervention effects must move the proximal mediator before the full answer.



Common-evidence readout control



Claim gates: route $\Rightarrow$ identity/coverage; state $\Rightarrow$ future retrieval/stop; terminal $\Rightarrow$ answer.

Matched-position, random-head, orthogonal, resample, and sham controls; full sequence endpoint.